SAMPLING AND QUANTIZATION ANALYSIS

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Introduction

This laboratory experiment focuses on the principles of sampling and quantization through MATLAB simulations. It illustrates how a continuous-time signal can be sampled, quantized, and reconstructed, with corresponding plots and spectral analyses to visualize each stage. The report also revisits the theoretical basis of the sampling theorem, first proposed by Nyquist and later advanced by Shannon, emphasizing its central role in digital communication systems.

Theoretical Framework

Before the advent of digital communication, information was primarily transmitted by modulating continuous carrier waves. In 1928, Harry Nyquist introduced the concept of distortion-free transmission, which became known as the Nyquist criterion. Claude Shannon later extended this idea, formally establishing the Nyquist–Shannon sampling theorem within the framework of noiseless communication channels.

Every communication system requires an information source such as speech, video, or other analog signals. Through sampling and quantization, these continuous signals are transformed into discrete-time sequences, making them suitable for digital processing. Mathematically, sampling converts a continuous function into a sequence of discrete values.

The Nyquist–Shannon theorem states:

A band-limited signal can be perfectly reconstructed from its samples if the sampling frequency is at least twice the highest frequency present in the signal.

This requirement prevents aliasing and ensures accurate reproduction of the original signal after sampling.

Objectives

- To verify the Nyquist Shannon sampling theorem using MATLAB simulations.
- To reconstruct the continuous signal from its sampled version.
- To apply quantization and observe its effects on the signal.

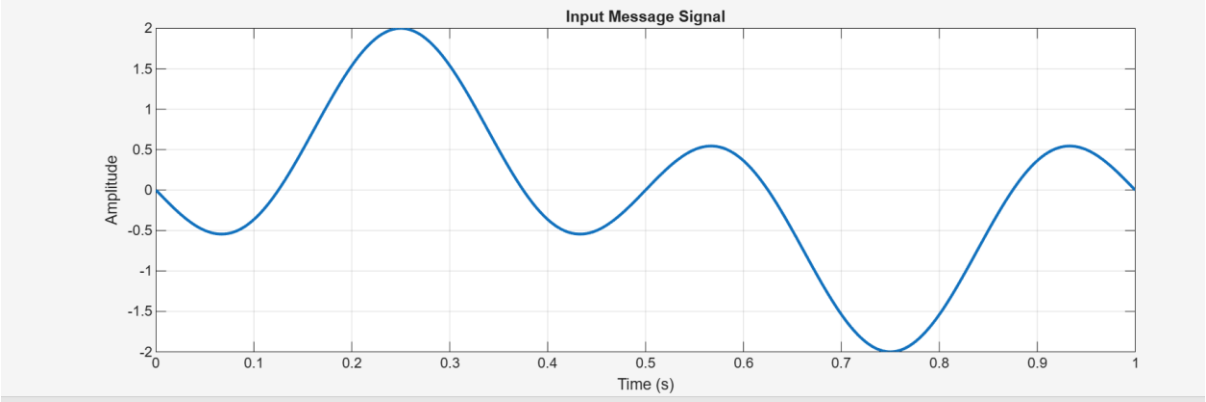
Apparatus

- MATLAB software
- Test message signal (sinusoidal waveform)

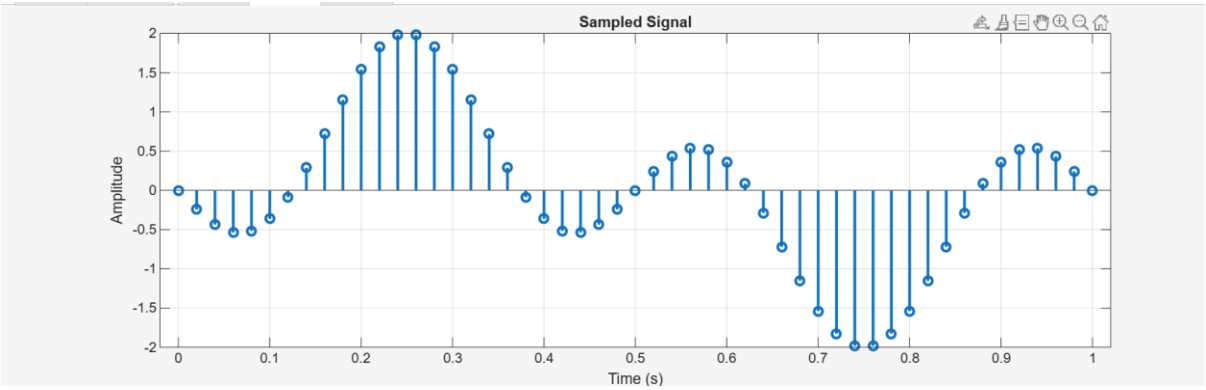
Results

The MATLAB simulations produced several plots, including:

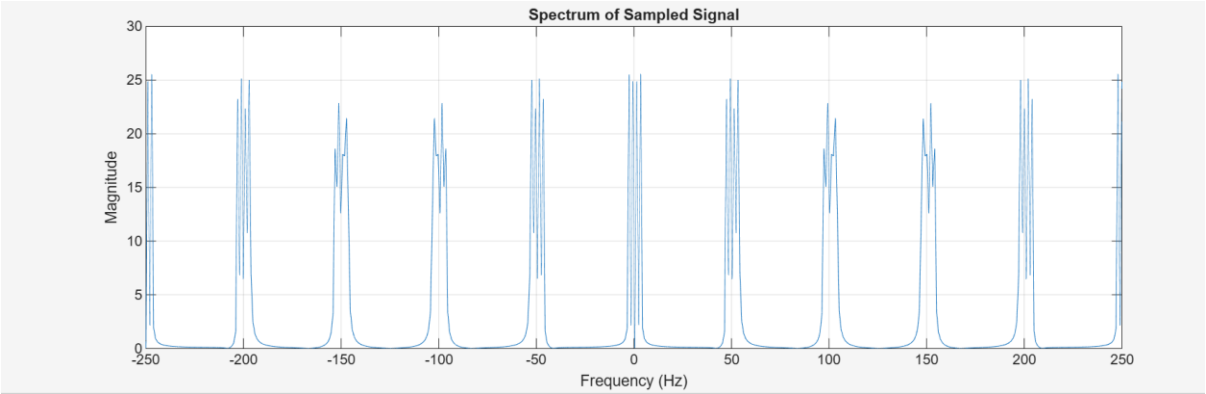
- The original continuous message signal



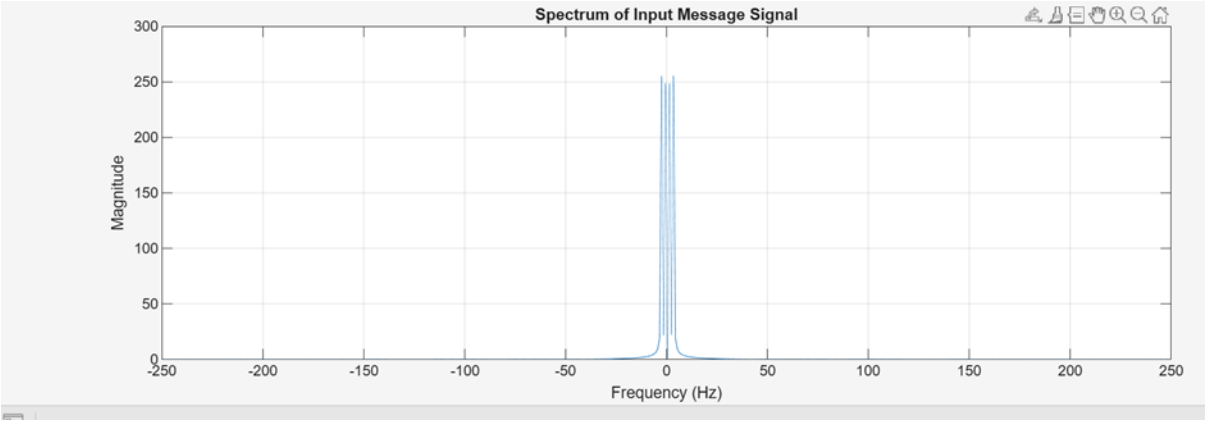
- The sampled signal in the time domain



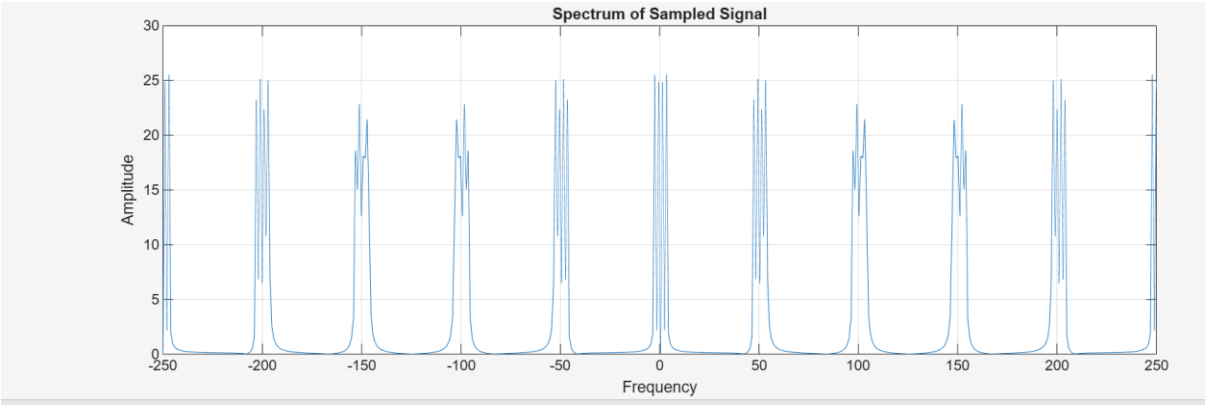
- Frequency spectrum of the sampled waveform



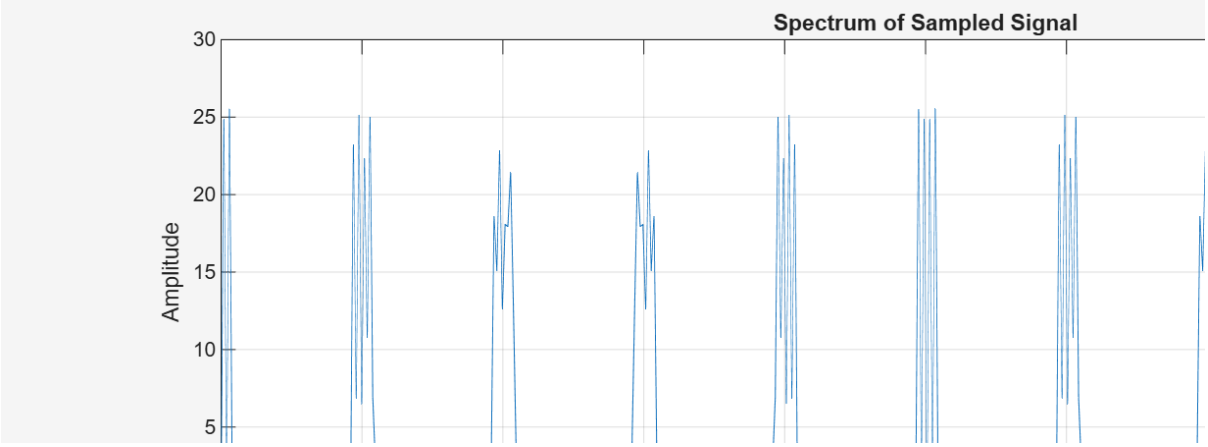
- Frequency spectrum of the input message signal



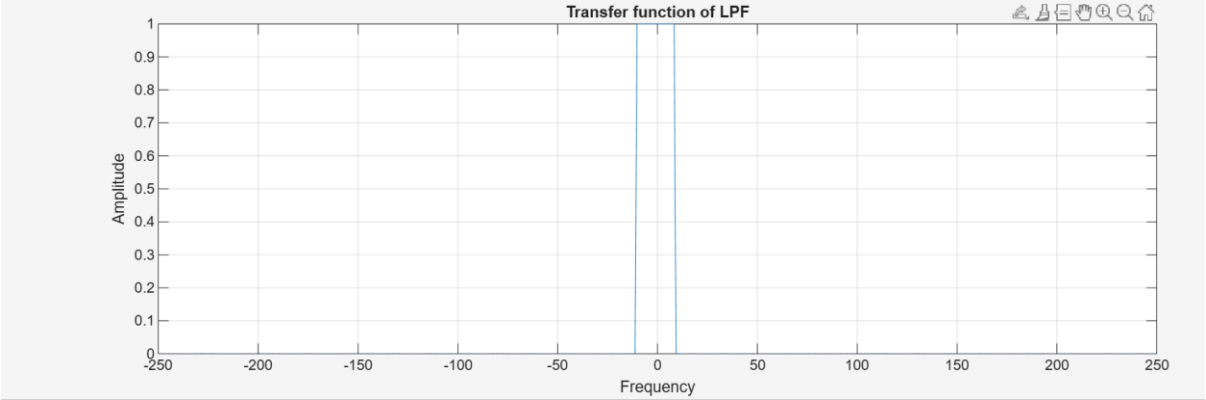
- Spectrum after low-pass filtering



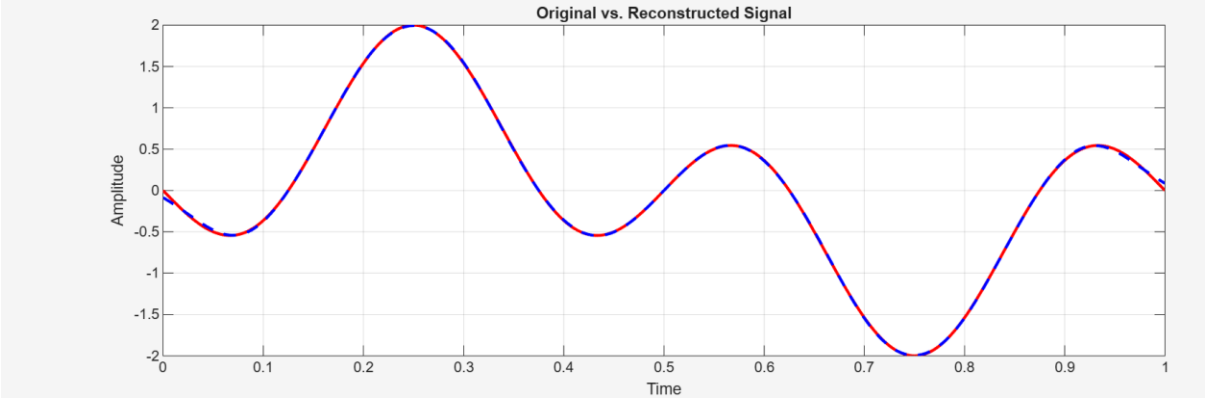
- A comparison of the reconstructed and original signals



- Transfer function of the applied LPF



Lastly, the original vs reconstructed signal



The sampled waveform closely resembled the original sinusoidal input, while the frequency-domain representation confirmed minimal distortion.

Discussion

The experimental results aligned well with the theoretical predictions of the sampling theorem. The input consisted of two sinusoidal signals at 1 Hz and 3 Hz, making 3 Hz the maximum frequency. Based on the Nyquist criterion, the minimum sampling frequency required was 6 Hz. In the simulation, however, a much higher sampling rate of 50 Hz was applied, ensuring distortion-free sampling.

FFT analysis of the sampled signal showed peaks at the expected frequencies, confirming that the spectral components were preserved. Low-pass filtering then eliminated higher-frequency replicas, successfully reconstructing the original waveform. The comparison between the reconstructed and original signals revealed negligible variation, validating the theorem in practice.

Possible sources of error include:

- Limited numerical precision in MATLAB, leading to minor truncation effects.
- Aliasing, if sampling were performed at a rate below twice the maximum frequency.

Conclusion

The MATLAB experiment effectively demonstrated sampling, quantization, and reconstruction, providing clear validation of the Nyquist–Shannon sampling theorem. The practical outcomes closely matched theoretical expectations, reaffirming the theorem's significance in digital communication. The objectives of the laboratory were fully achieved.

References

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